

Research Method

Spring-2026
Week 1 & 2 Lecture

Delivered by:
Prof. Dr. Faheem Akhtar Rajput
Chairman Dept. of AI & MMG

About Me

Name	Dr. Faheem Akhtar Rajput
Qualification	Ph.D. in Software Engineering (Machine learning – Data science)
Research Interests	Data Science, Feature Engineering, Machine Learning, Deep Learning, NLP, Image Classification, Bioinformatics
Publications	Above 60+ publications that include 30+ ISI indexed publications ,6 Book Chapter (Springer), International Indexed conferences. https://www.researchgate.net/profile/Faheem-Akhtar-3/research



Faheem Akhtar Rajputt

Edit

Ph.D (Data Science) · Associate Professor (BPS-20) at Sukkur IBA University
Sukkur, Pakistan

944.9 Research Interest Score | 1,329 Citations | 20 h-index

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A brief survey on breast cancer diagnostic with deep learning schemes using multi-image modalities

T Mahmood, J Li, Y Pei, F Akhtar, A Imran, KU Rehman
IEEE Access 8, 165779-165809

184

2020



MLPXSS: an integrated XSS-based attack detection scheme in web applications using multilayer perceptron technique

FMM Mokbal, W Dan, A Imran, L Jiuchuan, F Akhtar, W Xiaoxi
IEEE Access 7, 100567-100580

142

2019



Breast lesions classifications of mammographic images using a deep convolutional neural network-based approach

T Mahmood, J Li, Y Pei, F Akhtar, MU Rehman, SH Wasti
Plos one 17 (1), e0263126

114

2022



An intelligent fault detection approach based on reinforcement learning system in wireless sensor network

T Mahmood, J Li, Y Pei, F Akhtar, SA Butt, A Ditta, S Qureshi
The Journal of Supercomputing 78 (3), 3646-3675

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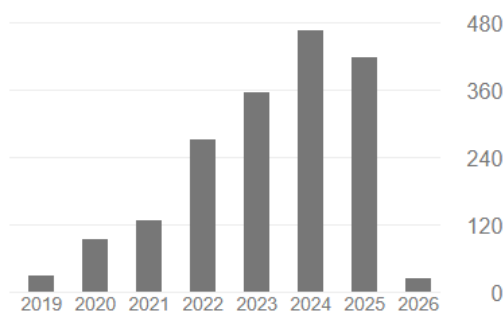
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About Me

Worked at Various Well-known Universities in Pakistan

Sukkur IBA University

NUCES-FAST Karachi

Muhammad Ali Jinnah University (MAJU)

Bahria University

And (various other universities as a visiting faculty).

Also worked in Industry as a Lead Trainer and Software Engineer

About Me

Awards – During PhD

Won Outstanding International Student Award (Ph.D.) from the **Ministry of Education China** in 2019.

Won **Outstanding Graduate Award (Ph.D.)** from Beijing University of Technology, China, in 2020.

Won **Special Contribution Award** for outstanding performance in curricular and extracurricular activities (Ph.D.) from **Beijing Municipal Government** and Beijing University of Technology, China.

Awarded with a **Certificate of Appreciation from the Embassy of Pakistan in Beijing** for outstanding Academic performance during my Ph.D. at Beijing University of technology china.

About Me

Editorial Affiliations

Guest Editor in [MDPI] **algorithms** Special Issue "AI Algorithms in Medical Imaging"

https://www.mdpi.com/journal/algorithms/special_issues/TB11D2U21N

Chairing 11th IEEE International Workshop on Medical Computing (MediComp) in collaboration with COMPSAC

(<https://ieeecompsac.computer.org/2023/medicomp-2023/>)

Chairing current and already Chaired 8th,9th and 10th and 11th IEEE International Workshop on Medical Computing (MediComp 2022) in collaboration with COMPSAC

(<https://ieeecompsac.computer.org/2022/>).

Part of various international conferences, especially the International Conference on Frontier Computing since 2017 as a Program Committee Member <https://www.fronticomp.com/>

Reviewer of various SCI/SCIE indexed Journals (IEEE Access, Journal of Supercomputing, and etc.).

Agenda

Course Introduction

Introduction to Research

Value of Research

Some basic terminology used in research and publications

Recap of the Lecture

Course Description

- This research methodology course focuses on developing researching and writing skills in Computer Science & Engineering Domain. As software development requires a multidisciplinary approach, many concepts are borrowed from social science, Psychology, Statistics, and other domains. Here we investigate the empirical research methods for their applicability and suitability to a research problem.

Course Objectives

- Students should be able:

	Course Learning Outcomes (CLOs)
1	Research Problem Analysis and Investigation: Students will be able to analyze complex computing-related research problems, conduct systematic literature reviews, and identify gaps in knowledge using appropriate empirical and investigative methods from computer science and related disciplines.
2	Design and Development of Research Solutions: Students will be able to design, develop, and evaluate research methodologies, tools, and experiments to address software-related research problems
3	Ethical and Professional Research Communication: Students will be able to produce well-structured, ethically sound research proposals, papers, and presentations that communicate findings effectively to academic and professional audiences, adhering to publishing standards and research ethics.

Where this course leads us?



Teaching Methodology

- 1. Theoretical Concepts followed by exercise..

Grading Criteria

Mid Term	30%
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Final	40%
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Research Paper	20%
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Quizzes / Assignments / Class Participation	10%
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Reading Material

Text Book

Booth, W. C., Booth, W. C., Colomb, G. G., Colomb, G. G., Williams, J. M., & Williams, J. M. (2008). *The craft of research*. University of Chicago press.

Reference Material

1. Olivier, M. S. (2009). *Information technology research: A practical guide for computer science and informatics*. Van Schaik.
2. Creswell, J. W., & Creswell, J. D. (2017). *Research design: Qualitative, quantitative, and mixed methods approaches*. Sage publications.
3. S.S. Vinod Chandra & S. Anand Hareendran. *Research Methodology*. Pearson publications.

Accessing the Course Material

- We will use LMS/Google Class Room
- Course Name on LMS/Google Class Room:
 Spring2026-F23-AI-RM-A
 Spring2026-F23-AI-RM-B
- Access Material using LMS/Google Class Room

What is Research?

What is Research

- Research is simply the process of finding solutions to a problem after thorough study and analysis of the situational factors.
- It is gathering information needed to answer a question, and thereby help in solving a problem.
- We do not do study in any haphazard manner. Alternatively, it is a systematic and organized effort to investigate.
- It is all the more necessary in case we want to repeat the study, or somebody else wants to verify our findings.
- Hence not only we have to do the study in a systematic manner but also that system should be known to others.

What is the Value of Research

- The nature of research problems could vary.
- Problems may refer
 - **to some undesirable situation**. For instance, in last few years Sindh Government announced 4000 jobs for PS, JS and HS teachers. More than 60,000 applicants applied for jobs. The test was conducted by STS, SIBAU. However, only about 2000 candidate could pass the test. Hence, one can investigate such undesirable situation. The researcher may come up with a variety of reasons that may relate with the applicants, the teachers, the curricula, the STS test, etc. In the same test, it was observed that the female applicants secured highest marks than male applicants. Thus, this situation can also be investigated as a separate research problem. Conducting such research offers the pleasure of solving a puzzle.
 - or these may refer to simply a **curiosity of the researcher** that may be agitating his or her mind. For instance, Newton's law of gravity.

What is the Value of Research

- Such findings make a good contribution to the body of knowledge i.e. making some good discoveries as part of the basic research. Finding answer to any enigma is self-satisfying.
- The researchers try to make use of their findings for **generating theories and models** that could be used for understanding human behavior and the functioning of different structures both at the micro (organizational) and macro (societal) level.
- Therefore, research may be considered as an *organized, systematic, data based, critical, objective, scientific inquiry or investigation into a specific problem*, undertaken with the purpose of finding answers or solutions to it.
- The value of research for policy makers, planners, business managers, and other stakeholders is that it reduces uncertainty by providing information that improves the decision-making process.



Basic Terminologies used in Research and Publications

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What is a research paper?

What is a research paper

- A research paper contains **original research results** or **reviewing existing results**. It is also called as academic paper or scholarly paper.

Types of Research Papers

Technical Paper

Review Paper

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- ☐ 1. **A secure versatile light payment system based on blockchain**
By: Zhong, Lin; Wu, Qianhong; Xie, Jan; et al.
FUTURE GENERATION COMPUTER SYSTEMS-THE INTERNATIONAL JOURNAL
Pages: 327-337 Published: APR 2019
[Full Text from Publisher](#) [View Abstract](#) 
- ☐ 2. **A Hitchhiker's Guide to the Blockchain Universe**
By: Waldo, Jim
COMMUNICATIONS OF THE ACM Volume: 62 Issue: 3 Pages: 38-42 Published: 2019
[Full Text from Publisher](#)
- ☐ 3. **Blockchain Electricity Trading Under Demurrage**
By: Devine, Mel T.; Cuffe, Paul
IEEE TRANSACTIONS ON SMART GRID Volume: 10 Issue: 2 Pages: 2323-2332
[Full Text from Publisher](#) [View Abstract](#) 
- ☐ 4. **Designing a Blockchain-Based IoT With Ethereum, Swarm, and LoRa availability with minimal security risks**
By: Ozyilmaz, Kazim Rifat; Yurdakul, Arda
IEEE CONSUMER ELECTRONICS MAGAZINE Volume: 8 Issue: 2 Pages: 21-28
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Where we can submit a research paper
for publication?

Where we can submit a research paper for publication?

- The research paper can be submitted for publication either in **conference** or in **journal**
- In general, the journal paper has more weightage than the conference paper.
- However, there are few conferences in each domain which have also high impact. For instance, **ICMLA**, **ICDM**, **MLDM** in **machine learning and Data Mining**.

Where we can submit a research paper for publication?

- Akhtar, Faheem, et al. "Diagnosis of large-for-gestational-age infants using a semi-supervised feature learned from expert and data." *Multimedia Tools and Applications* 79 (2020): 34047-34077.
- Akhtar F, Li J, Pei Y, Siraj S, Shaukat Z. Macrosomia fetus prediction with cluster-based feature selection scheme. In *Frontier Computing: Theory, Technologies and Applications (FC 2019)* 2020 Feb 26 (pp. 55-62). Singapore: Springer Singapore.
- Akhtar, Faheem, Jianqiang Li, Zahid Hussain Khand, Yu-Chih Wei, Khalid Hussain, and Sana Fatima. "An Efficient Small for Gestational Age Prognosis System Using Stacked Generalization Scheme (SGS)." In *2022 IEEE 46th Annual Computers, Software, and Applications Conference (COMPSAC)*, pp. 1456-1461. IEEE, 2022.

Diagnosis of large-for-gestational-age infants using a semi-supervised feature learned from expert and data

Faheem Akhtar^{1,2,3} · Jianqiang Li^{1,2} · Yan Pei⁴ · Azhar Imran¹ · Asif Rajput³ · Muhammad Azeem¹ · Bo Liu¹

Received: 29 August 2019 / Revised: 2 May 2020 / Accepted: 15 May 2020 /
Published online: 10 June 2020
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Abstract

In recent years, a rapid rise in the incidence of Large for gestational age (LGA) neonate is reported, and health care professionals employed themselves to discover the cause. Utmost of the previous studies were cohort or observational studies that employed simple linear or multivariate regression models, and very few of them employed machine learning (ML) schemes. Therefore, this research proposes to use 1 expert-driven and 7 automated feature selection schemes with well-known ML classifiers using 10 and 30 folds cross-validation. The induced results were compared with existing baselines, and Wilcoxon signed-rank test and the Friedman test were also introduced to verify the results. The ranked 20 features of the proposed expert-driven feature selection scheme outperformed amongst 7 automated feature selection schemes with a prediction precision, accuracy, and AUC scores of 0.94606, 0.84529, and 0.86492, respectively. Out of twenty features, eleven features were found similar to twenty ranked features of the automated feature selection schemes subsets. The classification results of the extracted features were utmost identical to the results of twenty features subset proposed by the expert-driven feature selection scheme. Therefore, we suggest pediatricians to refresh LGA diagnosis process with the proposed scheme because of its practical usage and maximum expert involvement.

Keywords Large for gestational age · Feature selection and extraction · Learning system · Prediction model · Machine learning

1 Introduction

Large for gestational age (LGA), also termed as Macrosomia, represents an abnormal fetus with excessive birth weight or a fetus with a birth weight above the 90th percentile of the birth weight of a similar fetus with the same gestational age and sex [5]. Neonates born as LGA are prone to adverse neonatal and maternal complications [14, 18].

✉ Yan Pei
peiyan@u-aizu.ac.jp

Extended author information available on the last page of the article.

Article

Diagnosis and Prediction of Large-For-Gestational-Age Fetus Using the Stacked Generalization Method

Faheem Akhtar^{1,2}, Jianqiang Li^{1,*}, Yan Pei³, Azhar Imran¹, Asif Rajput², Muhammad Azeem¹ and Qing Wang⁴

- ¹ School of Software Engineering, Beijing University of Technology, Beijing Engineering Research Center for IoT Software and Systems, Beijing 100124, China; fahim.akhtar@iba-suk.edu.pk (F.A.); azharimran63@gmail.com (A.I.); mazeem.qau@hotmail.com (M.A.)
 - ² Department of Computer Science, Sukkur IBA University, Sukkur 65200, Pakistan; asifali@iba-suk.edu.pk
 - ³ School of Computer Science and Engineering, the University of Aizu, Aizuwakamatsu 965-8580, Japan; peiyan@u-aizu.ac.jp
 - ⁴ Tsinghua National Laboratory for Information Science and Technology, Tsinghua University, Beijing 100084, China; qing.wang@tsinghua.edu.cn
- * Correspondence: lijianqiang@bjut.edu.cn; Tel.: +8610-67396842

Received: 30 August 2019; Accepted: 10 October 2019; Published: 14 October 2019

Abstract: An accurate and efficient Large-for-Gestational-Age (LGA) classification system is developed to classify a fetus as LGA or non-LGA, which has the potential to assist paediatricians and experts in establishing a state-of-the-art LGA prognosis process. The performance of the proposed scheme is validated by using LGA dataset collected from the National Pre-Pregnancy and Examination Program of China (2010–2013). A master feature vector is created to establish primarily data pre-processing, which includes a features' discretization process and the entertainment of missing values and data imbalance issues. A principal feature vector is formed using GridSearch-based Recursive Feature Elimination with Cross-Validation (RFECV) + Information Gain (IG) feature selection scheme followed by stacking to select, rank, and extract significant features from the LGA dataset. Based on the proposed scheme, different features subset are identified and provided to four different machine learning (ML) classifiers. The proposed GridSearch-based RFECV+IG feature selection scheme with stacking using SVM (linear kernel) best suits the said classification process followed by SVM (RBF kernel) and LR classifiers. The Decision Tree (DT) classifier is not suggested because of its low performance. The highest prediction precision, recall, accuracy, Area Under the Curve (AUC), specificity, and F1 scores of 0.92, 0.87, 0.92, 0.95, 0.95, and 0.89 are achieved with SVM (linear kernel) classifier using top ten principal features subset, which is, in fact higher than the baselines methods. Moreover, almost every classification scheme best performed with ten principal feature subsets. Therefore, the proposed scheme has the potential to establish an efficient LGA prognosis process using gestational parameters, which can assist paediatricians and experts to improve the health of a newborn using computer aided-diagnostic system.

Keywords: recursive feature elimination with cross-validation; stacked generalization; large for gestational age; feature engineering; machine learning; bioinformatics

1. Introduction

During the last several decades, an increased in the incidence of LGA neonates in the developed countries are reported, which is even wider in developing countries [1,2]. It is defined as a fetus whose gestational weight is above the 90th percentile of a fetus with a similar gestational age and sex [3]. It exhibits serious pre/post maternal complications, which are comprised of shoulder dystocia [4,5],

Article

An Automatic Detection and Localization of Mammographic Microcalcifications ROI with Multi-Scale Features Using the Radiomics Analysis Approach

Tariq Mahmood^{1,2}, Jianqiang Li^{1,3}, Yan Pei^{4,*}, Faheem Akhtar⁵, Azhar Imran⁶ and Muhammad Yaqub¹

- ¹ The School of Software Engineering, Beijing University of Technology, Beijing 100024, China; tmsheerazi@ue.edu.pk (T.M.); lijianqiang@bjut.edu.cn (J.L.); myaqubciitsw@gmail.com (M.Y.)
 - ² Division of Science and Technology, University of Education, Lahore 54000, Pakistan
 - ³ Beijing Engineering Research Center for IoT Software and Systems, Beijing 100124, China
 - ⁴ Computer Science Division, University of Aizu, Aizuwakamatsu, Fukushima 965-8580, Japan
 - ⁵ Department of Computer Science, Sukkur IBA University, Sukkur 65200, Pakistan; fahim.akhtar@iba-suk.edu.pk
 - ⁶ Department of Creative Technologies, Air University, Islamabad 44000, Pakistan; azharimran@mail.au.edu.pk
- * Correspondence: peiyan@u-aizu.ac.jp



Citation: Mahmood, T.; Li, J.; Pei, Y.; Akhtar, F.; Mudassir, A.; Yaqub, M. An Automatic Detection and Localization of Mammographic Microcalcifications ROI with Multi-Scale Features Using Radiomics Analysis Approach. *Cancers* **2021**, *13*, 5916. <https://doi.org/10.3390/cancers13235916>

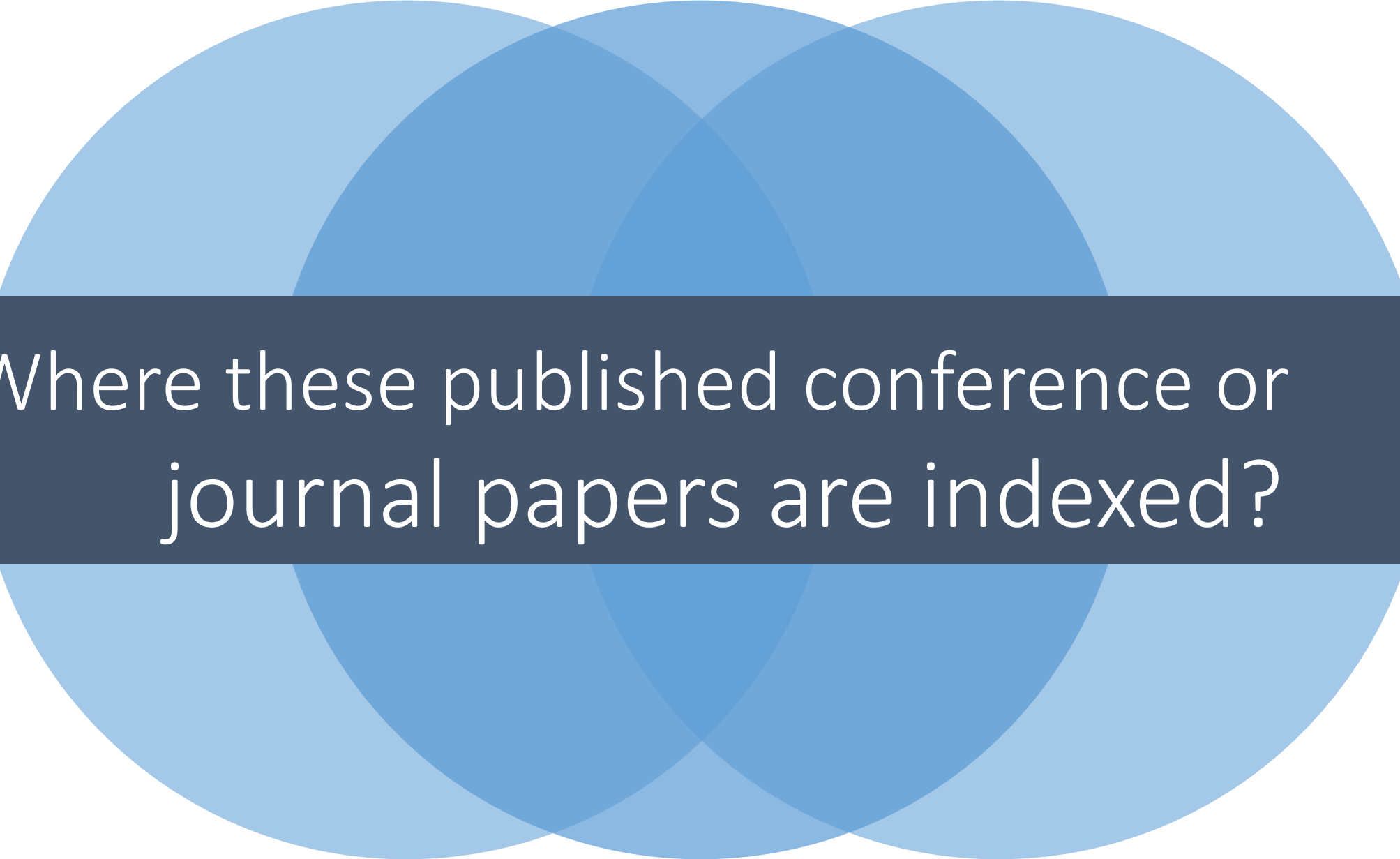
Academic Editors: Keyvan Farahani and Bardia Yousefi

Received: 23 September 2021
Accepted: 18 November 2021
Published: 24 November 2021

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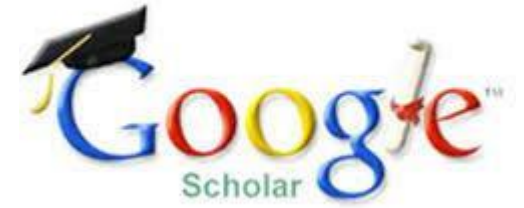
Simple Summary: Breast cancer is one of the foremost causes of cancer-related mortality in women. It is curable and controllable only if detected early. Microcalcifications in breast tissue are essential predictors for radiologists to detect early-stage breast cancer. This study proposes a method for detecting and classifying microcalcifications in mammogram images to predict breast lesions, using machine learning coupled with an interpretable radiomics approach. The method was evaluated using a publicly accessible dataset, which may aid radiologists and clinicians in identifying breast cancer in their regular clinical practices. This study contributes to the field of predictive modeling in healthcare.

Abstract: Microcalcifications in breast tissue can be an early sign of breast cancer, and play a crucial role in breast cancer screening. This study proposes a radiomics approach based on advanced machine learning algorithms for diagnosing pathological microcalcifications in mammogram images and provides radiologists with a valuable decision support system (in regard to diagnosing patients). An adaptive enhancement method based on the contourlet transform is proposed to enhance microcalcifications and effectively suppress background and noise. Textural and statistical features are extracted from each wavelet layer's high-frequency coefficients to detect microcalcification regions. The top-hat morphological operator and wavelet transform segment microcalcifications, implying their exact locations. Finally, the proposed radiomic fusion algorithm is employed to classify the selected features into benign and malignant. The proposed model's diagnostic performance was evaluated on the MIAS dataset and compared with traditional machine learning models, such as the support vector machine, K-nearest neighbor, and random forest, using different evaluation parameters. Our proposed approach outperformed existing models in diagnosing microcalcification by achieving an 0.90 area under the curve, 0.98 sensitivity, and 0.98 accuracy. The experimental findings concur with expert observations, indicating that the proposed approach is most effective and

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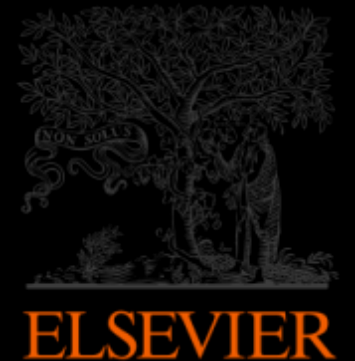
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Academic Databases



Academic Databases

- A conference or journal can be indexed in more than one databases (depends upon the subscription policy)
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- Publication is the responsibility of the publishers. Example: Elsevier, Springer, etc.



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ADVANCED SEARCH

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Diagnosis of Large for Gestational Age Fetus with an Expert-Driven Feature Selection Scheme

Publisher: IEEE [Cite This](#) [PDF](#)

Faheem Akhtar ; Jianqiang Li ; Yan Pei ; Bo Liu ; Muhammad Azeem ; Qing Wang ; Ji-Jiang Yang [All Authors](#)

2
Paper
Citations

54
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Text Views

Abstract

Document Sections

I. Introduction

II. Material and Methods

III. Results and Discussions

Abstract:

We propose to use an expert-driven feature selection scheme to diagnose and predict Large for Gestational Age (LGA) fetuses. A Fetus with excessive birth weight exhibits adverse neonatal and maternal complications. Early intervention can slow progression and prevent the upcoming complication of the disease. In this research, four well-known machine learning classifiers with ten-folds cross-validations are used to authenticate the proposed scheme. A Master feature vector is created, and an expert-driven feature selection scheme is proposed, which is later compared with existing published researches, master feature file created, and with an automated feature selection scheme. The best performance metrics (precision and AUC) scores are produced by random forest and logistic regression classifiers with the proposed expert-driven feature

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What is Impact Factor and ISI- Indexed Paper

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		2012 =123			2012 =19
		Sum: 181			Sum: 40
Calculation=	Cites to recent items	181	= 4.525		
	Number of recent items	40			



Dr. Faheem Akhter Rajput

Department of Computer Science, Sukkur IBA University, Pakistan

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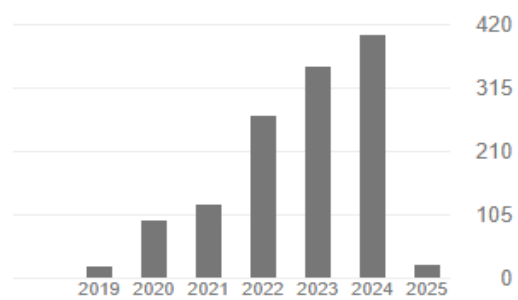
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T Mahmood, J Li, Y Pei, F Akhtar, A Imran, KU Rehman IEEE Access 8, 165779-165809				
☐ MLPXSS: an integrated XSS-based attack detection scheme in web applications using multilayer perceptron technique			119	2019
FMM Mokbal, W Dan, A Imran, L Jiuchuan, F Akhtar, W Xiaoxi IEEE Access 7, 100567-100580				



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3. CHEMISTRY, MULTIDISCIPLINARY	SCIE	148	2,195,290	1.401	5.222
4. CELL & TISSUE ENGINEERING	SCIE	18	76,359	3.535	4.940
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6. NEUROIMAGING	SCIE	13	118,110	2.374	4.732
7. GENETICS & HEREDITY	SCIE	164	994,322	2.505	4.467
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- If your paper is published in a journal which is available in **Master Journal List (MJL) AND Journal Citation Report (JCR)** then that journal is Impact factor and ISI-Indexed journal.
- Usually Impact factor and ISI-Indexed journals have coverage of **Science Citation Index Expanded**

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Journal Search

Search Terms

Search Type

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InCites™ Journal Citation Reports®

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Home

Go to Journal Profile

Master Search

Select Journals

Select Categories

Select JCR Year

2013

Select Edition

☒ SCIE ☒ SSCI

Journals By Rank

Categories By Rank

All Journal Categories ranked by Number of Journals

Show Visualization

1 - 25 of 232

Category	Edition	#Journals	Total Cites	Median Impact Factor	Aggregate Impact Factor
1. MULTIDISCIPLINARY SCIENCES	SCIE	46	2,879,971	0.798	5.880
2. CELL BIOLOGY	SCIE	165	1,784,283	3.333	5.815
3. CHEMISTRY, MULTIDISCIPLINARY	SCIE	148	2,795,290	1.401	5.222
4. CELL & TISSUE ENGINEERING	SCIE	18	76,359	3.535	4.940
5. NANO SCIENCE & NANOTECHNOLOGY	SCIE	73	799,992	1.768	4.902
6. NEUROIMAGING	SCIE	13	118,110	2.374	4.732
7. GENETICS & HEREDITY	SCIE	164	994,322	2.505	4.467
8. ASTRONOMY & ASTROPHYSICS	SCIE	59	781,855	1.676	4.462
9. EVOLUTIONARY BIOLOGY	SCIE	46	355,391	2.887	4.439
10. MICROBIOLOGY	SCIE	60	676,767	2.446	4.413

What is ISI-Indexed paper

- If your paper is published in a journal which is **not available in Master Journal List (MJL) BUT available in Journal Citation Report (JCR)** then that journal ISI-Indexed journal but has no any impact factor.
- Usually such journals have coverage of **Expanded Social Science Citation Index**

Tiers of ISI-
Indexed and
Impact
Factor
Journals

Tier-I or Quartile-I or Q1

Tier-II or Quartile-II or Q2

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JCR® Category	Rank in Category	Quartile in Category
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ENGINEERING, ELECTRICAL & ELECTRONIC	42 of 260	Q1
OPERATIONS RESEARCH & MANAGEMENT SCIENCE	8 of 84	Q1

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COMPUTER SCIENCE, INTERDISCIPLINARY APPLICATIONS	28 of 105	Q2
MEDICAL INFORMATICS	7 of 25	Q2

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Research Domain

Computer Science

Medical Informatics

2.

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3.

JOURNAL OF BIOMEDICAL INFORMATICS

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Recap of the Lecture

- Course Introduction
- Introduction to Research
- Value of Research
- Some basic terminology used in research and publications